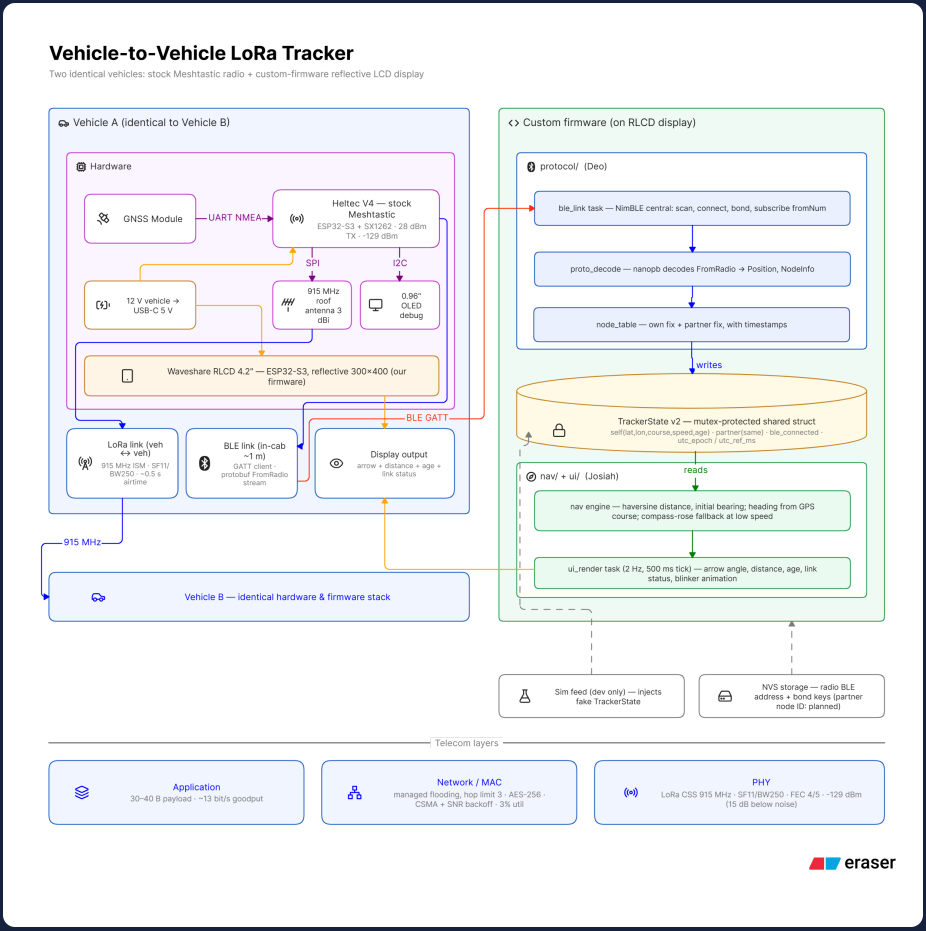
OFF-GRID VEHICLE TRACKER

Technical Manual

Architecture, radio physics, firmware, and the complete failure catalog

This manual presents the system architecture, the radio-frequency design with full derivations and worked numbers, the firmware, the navigation mathematics, the provisioning and verification procedures, and the complete catalog of failure modes learned on real hardware. It is written so that an engineer who has never seen this system can operate, debug, extend, and rebuild it.



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1. System overview and the five ideas

Two Heltec WiFi LoRa 32 V4 radios (callsigns SF and RG) run unmodified Meshtastic firmware 2.7.26 and exchange position, telemetry, and text over a private 915 MHz LoRa channel. Each radio serves one Waveshare ESP32-S3-RLCD-4.2 driver display running our custom firmware, connected over Bluetooth Low Energy using the Meshtastic Client API [2]. The displays decode the radio traffic into a live navigation UI. There is no phone, tower, or internet dependency anywhere in the chain.

Five ideas explain essentially every behavior and every failure this system has ever shown. They recur throughout this manual.

> **IDEA 1. AGREEMENT IS THE CONNECTION.** LoRa has no pairing or session. Two radios exchange data exactly when they independently agree on frequency, modulation, and encryption key. Every "total silence" failure in this project was one of those three quietly disagreeing.

IDEA 2. THE NAME IS THE FREQUENCY. Meshtastic derives the operating frequency from a hash of the primary channel name [2]. Rename the channel and the radio retunes. The OLED LoRa screen shows the synthesized frequency and is the ground truth; configuration read-backs can lie (Section 12, entries 1 and 6).

IDEA 3. DECIBELS ARE RATIOS, AND LORA DECODES BELOW THE NOISE. Signal quantities are logarithmic: +3 dB is double, dBm anchors to 1 mW. At spreading factor 11 the receiver decodes down to 17.5 dB below the noise floor [1], because each bit is spread over 2048 chips that integrate coherently.

IDEA 4. ONLINE IS ARITHMETIC, NOT A CONNECTION. Every liveness indicator is a comparison between now and a stored last-heard timestamp [3]. Every counter lie we chased was a timestamp problem, never a radio problem.

IDEA 5. AIRTIME IS THE BUDGET. Each packet costs roughly half a second of channel time (Section 3.3); the firmware throttles position broadcasts above 25 percent utilization [2]. Design every periodic packet to name the failure it prevents.

2. Hardware inventory and identification

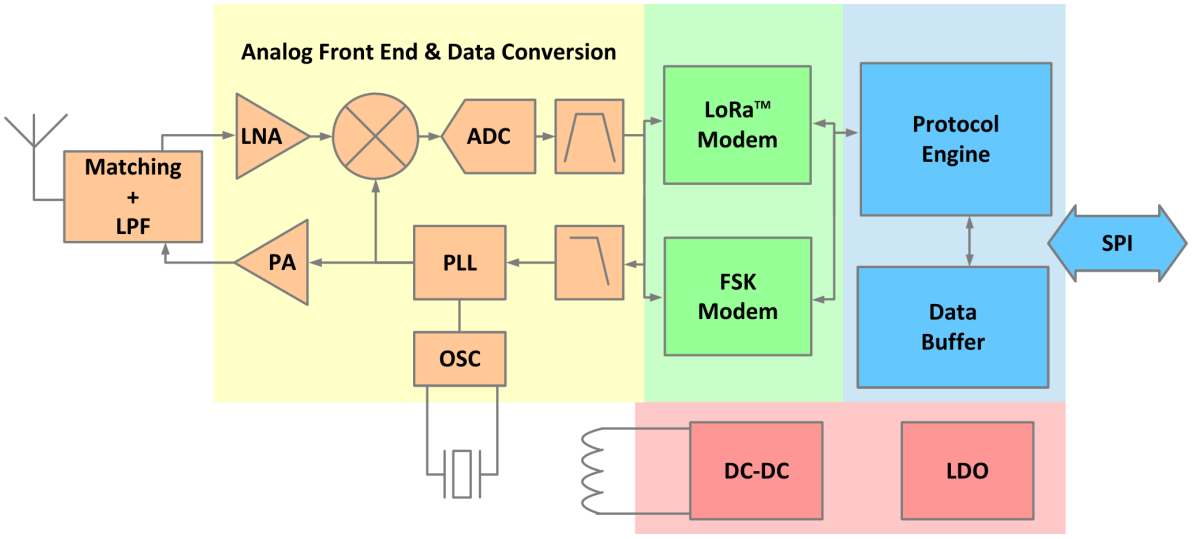
Unit	Identity	Key facts
Radio SF	Node `!49b5ead4`, BLE MAC `8c:fd:49:b5:ea:d5`	Heltec V4: ESP32-S3 MCU, Semtech SX1262 transceiver [1], u-blox GNSS, OLED status screen, lithium cell. Pairs with Display 1. `bluetooth.mode NO_PIN`.
Radio RG	Node `!49b5bb5c`, BLE MAC `8c:fd:49:b5:bb:5d`	Same hardware. Pairs with Display 2. `bluetooth.mode NO_PIN`.
Display 1	USB serial `A4:CB:8F:D0:7D:40`	Waveshare ESP32-S3-RLCD-4.2: reflective LCD 400x300, sunlight readable, no backlight; runs our firmware.
Display 2	USB serial `94:A9:90:DD:14:3C`	Identical firmware; remembers RG in NVS after its first scan.

Flashing trap: radios and displays share USB VID:PID 303A:1001. Radio MACs begin `8C:FD:49`. Always check the `SER=` field of the COM port before any upload. Flashing a radio with display firmware erases its Meshtastic and requires factory recovery (Section 12, entry 8).

Enclosures: radios ride in community-printed cases; the display case is the AWSW "Symmetrical case" (Printables model 1610004, CC BY-NC 4.0) [6], printed in PETG for dashboard heat tolerance, fastened with four M2.5x6 screws. Six millimeters maximum: longer screws press on the panel glass.

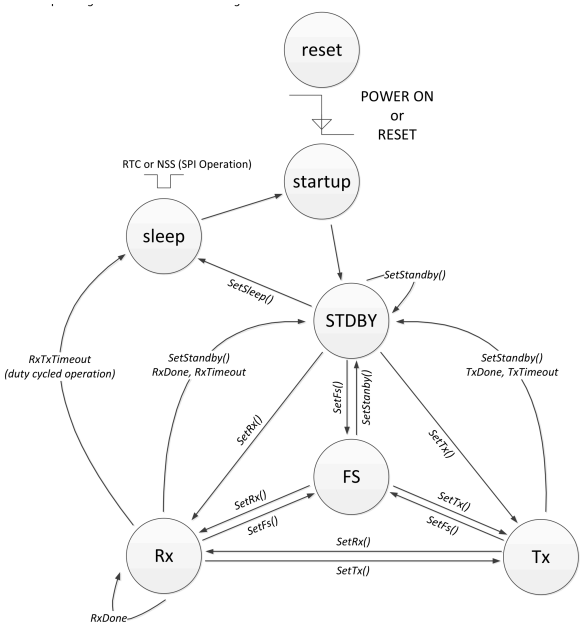
2.1 Inside the transceiver

The SX1262's internal block diagram repays a careful read, because the entire physical layer of Section 3 lives inside it. On receive, the antenna signal passes through the matching network into a low-noise amplifier, is mixed down from 913 MHz by the phase-locked loop, digitized, and handed to the LoRa modem; the protocol engine assembles packets into the data buffer, which the ESP32-S3 drains over SPI. The PLL matters beyond the signal path: when Meshtastic computes a frequency from the channel name (Section 3.2), this synthesizer is the hardware that gets programmed, and the OLED frequency readout this manual treats as ground truth is a report of that programming.



SX1261/2 internal block diagram, adapted from the datasheet rev 2.2, fig. 4-1 [1].

The transceiver is always in exactly one of the circuit modes below, and the diagram contains no state in which the chip listens and transmits at once. That single fact is the physical root of the airtime discipline in Section 4.2: a node spending 0.56 s transmitting is deaf for all of it, so simultaneous transmissions on one channel simply lose each other's packets. Section 4.3 explains how the protocol lives with that.



Transceiver circuit modes, adapted from the datasheet rev 2.2, fig. 9-1 [1].

3. RF design: the contract, the slot math, the modulation

3.1 The contract (Idea 1)

Layer	Our value	If mismatched
Frequency	Slot 43 = 912.875 MHz, derived from channel name "Group9"	Radios tuned apart; zero coupling
Modulation	LONG_FAST preset: SF11, 250 kHz bandwidth, coding rate 4:5	Chirps undecodable
Channel encryption	AES-256 pre-shared key, distributed in the channel URL	Packets heard, payloads dropped
Node identity	Curve25519 public keys exchanged via NodeInfo	Direct messages fail silently or NAK

3.2 Frequency slot derivation (Idea 2)

Meshtastic does not store a frequency; it computes one. The United States band plan runs from 902 to 928 MHz, which at a channel width of 250 kHz divides into 104 slots. The firmware hashes the primary channel name and takes the result modulo the slot count, so the operating frequency is a deterministic function of the name:

$$f_{slot} = f_{start} + \frac{BW}{2} + [\text{djb2}(name) \bmod N_{slots}] \cdot BW, \qquad N_{slots} = \frac{f_{stop} - f_{start}}{BW} = \frac{26\ \text{MHz}}{250\ \text{kHz}} = 104$$

The hash function is djb2, a simple string hash that starts from the constant 5381 and, for each character of the name, multiplies the running value by 33 and adds the character's code; the slot index is that value modulo 104. The formula can be calibrated against a known constant: djb2 of "LongFast" modulo 104 equals 19, which corresponds to 906.875 MHz, the published United States default [2]. Applying the same arithmetic to our channel name "Group9" yields slot 43, and substituting slot 43 into the slot formula gives the frequency that both of our radios must display:

$$f_{43} = 902.000 + 0.125 + 43 \times 0.250 = 912.875\ \text{MHz}$$

Consequence, proven the hard way: two radios with identical keys but different primary channel names sit megahertz apart and never couple. The public regional mesh lives on slot 19, so slot 43 carries only our traffic. The frequency is public airspace; privacy comes from the key (Section 5).

3.3 Modulation and time on air (Ideas 3 and 5)

LoRa is a chirp spread spectrum modulation: each symbol is a frequency sweep across the full channel, and the information is carried in the sweep's starting offset. A spreading factor of 11 provides $2^{11} = 2048$ possible offsets, so each symbol carries 11 bits. The duration of one symbol is the number of chips divided by the bandwidth:

$$T_{sym} = \frac{2^{SF}}{BW} = \frac{2^{11}}{250{,}000\ \text{Hz}} = 8.19\ \text{ms}$$

De-chirping integrates 2048 chips coherently, which is why the SX1262 demodulates at signal-to-noise ratios down to -17.5 dB at SF11 [1]. The coding rate of 4:5 adds one parity bit for every four data bits, which allows the receiver to repair short bursts of errors. The number of symbols needed to carry a payload follows Semtech's published formula [4], in which PL is the payload length in bytes, H is zero when an explicit header is present, DE is the low-data-rate optimization flag, and CR equals 1 for rate 4:5. The formula converts a byte count into a symbol count: the numerator counts the payload bits plus overhead, the denominator is the number of bits each symbol carries, and the ceiling and the factor of (CR + 4) account for forward error correction:

$$N_{payload} = 8 + \max\left(\left\lceil \frac{8PL - 4SF + 28 + 16\,CRC - 20H}{4(SF - 2DE)} \right\rceil (CR + 4),\ 0\right)$$

Worked example. Consider a typical 40-byte position packet, transmitted with SF = 11, BW = 250 kHz, DE = 0, the CRC enabled, and a 16-symbol preamble. Substituting these values gives 48 payload symbols, and adding the preamble and the 4.25 synchronization symbols gives the total time on air:

$$N_{payload} = 8 + \left\lceil \frac{8(40) - 44 + 28 + 16}{44} \right\rceil \times 5 = 48, \quad T_{pkt} = (16 + 4.25 + 48)\,T_{sym} = 68.25 \times 8.19\ \text{ms} \approx 0.56\ \text{s}$$

Each position update therefore occupies roughly 0.56 seconds of air. This single number drives the entire cadence and airtime design presented in Sections 4.2 and 9: it is the reason the system reports every ten seconds rather than every second.

4. Link budget and airtime budget

4.1 Link budget

A link budget adds every gain and subtracts every loss between the transmitter and the receiver, and then asks whether the power that arrives is still above the receiver's sensitivity. The received power is the transmitted power plus the two antenna gains minus the free-space path loss, where the path loss grows with both distance and frequency:

$$P_{rx} = P_{tx} + G_{tx} + G_{rx} - FSPL, \qquad FSPL = 20\log_{10} d + 20\log_{10} f - 147.55 \ \ (d \text{ in m},\ f \text{ in Hz})$$

The table below lists the value each term takes in this system and where that value comes from.

Term	Value	Source
Transmit power P_tx	+30 dBm (1 W)	Config; US legal maximum for this band [5]
Antenna gains G_tx, G_rx	~0 dBi each	Stock quarter-wave whips
Receiver sensitivity S	-131 dBm	SX1262 datasheet at SF11 / 250 kHz [1]
Total budget P_tx - S	161 dB	Loss the link can absorb before decode fails

Worked example. Evaluating the path loss formula at 913 MHz for distances of 1 km and 5 km shows how gently the loss grows with distance:

$$FSPL_{1\,\text{km}} = 60 + 179.2 - 147.55 = 91.7\ \text{dB} \Rightarrow P_{rx} = -61.7\ \text{dBm}; \qquad FSPL_{5\,\text{km}} = 105.7\ \text{dB} \Rightarrow P_{rx} = -75.7\ \text{dBm}$$

Even at 5 km the received signal of -75.7 dBm sits 55 dB above the receiver's sensitivity in free space. Streets are not free space, however: we reserve 20 dB of that headroom for terrain, buildings, and fading, which places the honest design range at 3 to 5 km. As a measured baseline, a round-trip traceroute at desk range reports 5.75 to 7.25 dB of signal-to-noise ratio in each direction. Spare margin converts into distance through the doubling rule, which follows from the logarithm in the path loss formula: every 6 dB of margin roughly doubles the achievable range:

$$\frac{d_2}{d_1} = 10^{\Delta/20} \Rightarrow \Delta = 6\ \text{dB} \approx \times 2\ \text{distance}, \qquad \text{margin} = SNR_{meas} - SNR_{floor} = 6 - (-17.5) = 23.5\ \text{dB}$$

4.2 Airtime budget (Idea 5)

Meshtastic suppresses position broadcasts once channel utilization rises above roughly 25 percent [2], so the design must show that normal operation stays below that throttle. The utilization is simply the fraction of each reporting period the channel spends carrying packets: the number of transmitting nodes multiplied by the time on air of one packet, divided by the reporting cadence. With two vehicles reporting a 0.56-second packet every ten seconds, the arithmetic works out as follows:

$$U = \frac{N_{nodes} \cdot T_{pkt}}{T_{cadence}} = \frac{2 \times 0.56}{10} \approx 11\%\ \text{(no relay)}; \quad \times 2\ \text{relayed} \approx 22\%\ \text{vs the 25\% throttle}$$

The default mesh role relays every overheard packet (hop limit 3); with exactly two nodes every relay is waste and doubles the spend to roughly 22 percent, brushing the throttle precisely when both vehicles move. Both radios therefore run role `CLIENT_MUTE` (receive and send everything, relay nothing), measured to halve idle utilization from 9-13 percent to 5-6 percent. Revert both to `CLIENT` if a third node or repeater ever joins.

4.3 Collision handling: who talks first

The half-duplex state machine of Section 2.1 raises the obvious edge case: both radios boot together and transmit their NodeInfo at the same instant. Nobody is assigned to talk first, and that is deliberate; the design philosophy is the classic ALOHA one, do not try to prevent every collision, make collisions cheap, detectable, and non-repeating. Four mechanisms stack. First, listen before talk: before transmitting, each radio runs the SX1262's channel activity detection, and whichever radio keyed up even milliseconds earlier wins the channel while the other hears its preamble and defers. Second, random backoff: in a true tie both packets are lost, nothing crashes, and each radio retries after a freshly drawn random delay, so a repeat collision has probability 1/N for an N-slot contention window, then 1/N squared, collapsing geometrically. Determinism would be the dangerous design here; two identical radios on a fixed schedule would collide forever. Third, repetition: directed packets are acknowledged and retransmitted, and broadcasts such as NodeInfo are periodic, so a lost boot announcement costs one cycle of patience. Fourth, physics: LoRa receivers exhibit capture, decoding the stronger of two overlapping transmissions when they differ by roughly 6 dB, so many theoretical collisions still deliver a packet [9]. Measured on this bench, two radios power-cycled together appear in each other's node lists within a minute, every time. The full analysis, with a timing diagram, is in `docs/COLLISION_AVOIDANCE.md`.

5. Security model

- **Channel traffic** (positions, broadcasts, canned texts): AES-256 with a 256-bit pre-shared key generated at channel creation and distributed only inside the provisioning URL over USB [2]. The key never crosses the air.
- **Direct messages and position requests**: additionally encrypted per node with Curve25519 public keys exchanged via NodeInfo. A stale stored key makes direct traffic fail one-way and silently (Section 12, entry 4).
- **What an eavesdropper on slot 43 learns**: packet headers are cleartext, so node IDs, timing, and signal levels are visible. Payloads are not. Existence and schedule leak; content and location do not.
- **Key custody**: the channel URL embedding the key lives in the private team repository by deliberate decision. Never publish it. Rotating to a fresh channel takes minutes and orphans the old key.

6. Radio configuration reference and provisioning

Setting	Value	Rationale
`channel (primary)`	Group9, position precision 32	Private slot 43; full-resolution positions (the channel default of 13 truncates every broadcast to a ~6 km grid)
`lora.region`	US	902 to 928 MHz plan [5]
`lora modem preset`	LONG_FAST	SF11 / 250 kHz / 4:5, the range versus rate compromise
`lora.tx_power`	30	Legal maximum; margin is range
`position.position_broadcast_secs`	60	Parked cadence with a fix
`position.broadcast_smart_minimum_interval_secs / _distance`	10 / 20	Moving cadence: 10 s AND 20 m must both pass
`position.gps_update_interval`	10	Must be ≤ the smart interval or stale fixes re-broadcast
`telemetry.device_update_interval`	300	Liveness floor; keeps counters honest at ~0.17 percent airtime
`device.role`	CLIENT_MUTE	Two-node mesh: no relaying (Section 4.2)
`bluetooth.mode`	NO_PIN	Just-Works pairing for the display; bonded-mode radios stop advertising and become unreachable
Canned messages	21 full-word presets	Hard firmware cap: the whole set is one string of at most 200 characters (ours uses 194)

Provisioning procedure

The command sequence below takes a radio from factory state to full team specification using the Meshtastic command-line tool over USB. The first command applies the channel URL, which carries the channel name, the encryption key, and the position precision in one string; the placeholder `<CHANNEL-URL>` stands for the actual URL, which is stored in `hardware/README.md` in the private team repository and must

never be published. The remaining commands apply the position cadence, the telemetry heartbeat, the mesh role, the Bluetooth mode, the radio's callsign, and the canned message list described in the table above.

```
# The channel URL (containing the key) is in hardware/README.md. Never publish it.
python -m meshtastic --port COMx --seturl "<CHANNEL-URL>"
python -m meshtastic --port COMx --set position.position_broadcast_secs 60 \
  --set position.broadcast_smart_minimum_interval_secs 10 \
  --set position.broadcast_smart_minimum_distance 20 \
  --set position.gps_update_interval 10 \
  --set telemetry.device_update_interval 300 \
  --set device.role CLIENT_MUTE
python -m meshtastic --port COMx --set bluetooth.mode NO_PIN     # radio that serves a dis
python -m meshtastic --port COMx --set-owner-short "SF"          # or "RG"
python -m meshtastic --port COMx --set-canned-message "OK|Yes|No|Hello|..."   # full set i
```

Verification is part of the procedure, not optional: read every value back with `--get`, confirm the OLED LoRa screen reads 912.875, then power-cycle and confirm 912.875 survives. An unsaved channel write is the classic silent revert (Section 12, entries 1 and 6).

7. Display firmware architecture

The display firmware lives in the PlatformIO project `firmware-display/`, built on the pioarduino platform with Arduino core 3.x. It comprises roughly 2200 hand-written lines of C and C++ plus protobuf decoding code generated by nanopb. The architecture is a one-directional dataflow with exactly one writer and one reader meeting at a mutex-protected store, which eliminates an entire class of concurrency bugs by construction. The diagram below traces the two tasks: the left column follows a radio packet from the Bluetooth link down into the shared store, and the right column follows the user-interface task as it reads the store and renders the screen twice per second.

```
BLE task (NimBLE central)                          UI task (2 Hz render)
 ble_link: scan/connect/pair/subscribe/drain        tracker_store_get()
   -> proto_decode (nanopb FromRadio)                 -> dead_reckon projection (Sec. 8)
   -> node_table (positions, names, batteries,        -> nav_compute (haversine, bearing)
      texts, timestamps)                              -> ui_render (3 pages on the RLCD)
   -> tracker_store_set(TrackerState v5)
```

7.1 The contract: TrackerState (version 5)

The header `include/tracker_state.h` defines the single data structure that crosses between the two tasks, shown below. It is co-owned by the protocol side and the user-interface side, it is versioned, and changes to it ship as one agreed commit. Because it is pure data with no operating-system or Arduino dependencies, it also compiles on a desktop computer, which is what makes the host-side unit tests of Section 10 possible. The `Fix` structure holds one vehicle's position solution; the outer structure then pairs the two fixes with the link status, the wall clock, the radio callsigns, the message history, and both battery levels.

```
typedef struct { double lat_deg, lon_deg; float course_deg, speed_mps;
                 uint32_t age_s; bool valid; } Fix;
typedef struct {
  Fix self, partner;            // own vehicle; partner vehicle
  bool ble_connected;           // in-cab link to our radio
  uint32_t utc_epoch, utc_ref_ms;          // GPS wall clock + millis reference
  char self_name[8], partner_name[8];      // radio callsigns from NodeInfo
  uint32_t msg_seq, msg_ms;                // latest received text (popup)
  char msg_from[8], msg_text[40];
  uint8_t msg_hist_n;                      // received-message history, newest first
  TrackerMsg msg_hist[4];
  uint8_t self_batt, partner_batt;         // 0..100; 101 external power; 255 unknown
} TrackerState;
```

7.2 BLE link (Meshtastic Client API [2])

GATT item	UUID	Role
Service	`6ba1b218-15a8-461f-9fa8-5dcae273eafd`	Meshtastic Client API
ToRadio	`f75c76d2-129e-4dad-a1dd-7866124401e7`	Write: commands and outbound packets
FromRadio	`2c55e69e-4993-11ed-b878-0242ac120002`	Read until empty: config dump and inbound packets (firmware 2.x UUID)
FromNum	`ed9da18c-a800-4f66-a670-aa7547e34453`	Notify: "new data queued", triggers a drain

On connect: write `want_config_id`, drain the configuration dump (node DB, MyInfo), then drain on every notify. The radio advertises only from boot until its first BLE disconnect, so the product flow is ignition-on: power the radio and display together; the display fires direct connect attempts and catches the boot window. The radio's address persists in NVS; a scan that finds any Meshtastic service re-learns it, which is how replacement units pair automatically.

7.3 Decode and node table

The decoder handles four Meshtastic port numbers. POSITION_APP carries positions; the speed arrives on the wire in km/h, the display stores it in m/s, and a 4.5 km/h deadband suppresses the jitter a parked GPS produces. NODEINFO_APP carries node names and public keys, TELEMETRY_APP carries battery levels, and TEXT_MESSAGE_APP carries raw UTF-8 text. Two honesty mechanisms live here: a bit-identical position re-broadcast does not reset the data age (a fixless radio re-sending its stored fix must not look fresh), and texts from our own node number are discarded because the radio echoes sent messages over BLE and only the receiving display should notify.

7.4 UI features

The interface consists of three pages: the compass, the status page, and the messages page. A short press of the button advances to the next page, while a long press resets the tracking state, or clears the message history when the messages page is showing. Every element is labeled with the radio callsigns learned from

NodeInfo. A link tag is drawn while the partner has been heard within the last 90 seconds. A received text appears as a popup for eight seconds on the receiving display only, and is retained in a four-entry history. The closing, apart, or holding trend label is computed as described in Section 8.4, and a banner warns when the partner's battery falls to 20 percent. Finally, the BOOT button on the display transmits a POSITION_APP want-response packet to the partner, an on-demand position request identical to the one the official phone application sends.

Two of these behaviors were redesigned after the 2026-10-04 two-vehicle night drive, and both changes are road-test lessons worth recording. First, the compass is **course-up while driving**: the top of the ring is always straight ahead, the cardinal letters rotate with the vehicle's course (heading south puts S at the top), the dart shows the partner relative to the hood, and a fixed index mark at twelve o'clock marks the nose. The original north-up ring forced the driver to physically rotate the unit to match the road, at which point nothing on it could be read. Stopped, with no course to be relative to, the ring falls back to north-up with the dart at absolute bearing, matching the parked "SE of you" wording. Second, the early-staleness cue (hollow dart plus inverted age) is **cadence-aware**: the threshold is 30 seconds while the partner's last reported speed says they are moving (three missed 10-second packets, genuinely suspicious) but relaxes to 90 seconds when they are parked, because a parked radio only speaks every 60 seconds and the drive test showed the cue alarming at 59 seconds of perfectly normal silence.

7.5 Operator-facing functions and what they do on the wire

The Device User Guide tells an operator which buttons to press; this section records what each action costs or changes at the protocol level, which is what an engineer needs when reasoning about airtime or debugging behavior. On the radio, **Send Position** transmits one immediate POSITION_APP broadcast (one 0.56 s packet, outside the smart-broadcast schedule). **Sleep Screen** only blanks the OLED; the radio keeps transmitting and receiving normally. **Mute Channel** suppresses local notifications only; packets are still received, decrypted, and forwarded to the display, so the partner's tracking is unaffected. **Reset NodeDB** erases stored node records including public keys; it is the prescribed cure for the stale-key failure of Section 12 entry 4 and is otherwise destructive of nothing but caches, since records repopulate from live traffic. The Clock and GPS Action menus change formatting and polling presentation only. On the display, the MAIN button's long press clears RAM-side tracking state (the radio's node database is untouched), the BOOT button transmits a directed POSITION_APP want-response packet (per-node encrypted, acknowledged, retried), and the RESET button is a hardware reset of the ESP32-S3; the BLE bond and the paired radio's address persist in NVS, so the display reconnects without re-pairing.

8. Navigation mathematics

All navigation outputs are computed on the display from the two stored fixes. Angles use latitude φ and longitude λ in radians; ψ is own course over ground.

8.1 Distance: haversine

The distance between the vehicles is the great-circle distance between their two fixes, computed with the haversine formula. The intermediate quantity a measures how far apart the two points are on the unit sphere, and the arcsine converts it back into an arc length along the Earth's mean radius R:

$$a = \sin^2\frac{\Delta\varphi}{2} + \cos\varphi_1 \cos\varphi_2 \sin^2\frac{\Delta\lambda}{2}, \qquad d = 2R \arcsin\sqrt{a}, \qquad R = 6371 \text{ km}$$

Worked example from the bench: RG at (33.58656, -101.87542), SF at (33.58646, -101.87536). Δφ = 0.00010°, Δλ = 0.00006°. The haversine gives d ≈ 12.5 m ≈ 41 ft, exactly what the status page showed. At this scale the formula degenerates gracefully to flat-earth Pythagoras; it is used because it stays accurate at every scale up to antipodes.

8.2 Bearing and the arrow

The direction from one vehicle to the other is the initial great-circle bearing, obtained from the two fixes with the standard two-argument arctangent form. The second expression then subtracts the driver's own course and wraps the result into the range of plus or minus 180 degrees, giving the angle the arrow must make relative to the vehicle's nose:

$$\theta = \operatorname{atan2}(\sin\Delta\lambda \cos\varphi_2,\ \cos\varphi_1 \sin\varphi_2 - \sin\varphi_1 \cos\varphi_2 \cos\Delta\lambda), \qquad \alpha_{arrow} = \operatorname{wrap}_{\pm 180}(\theta - \psi_{course})$$

θ is the initial great-circle bearing to the partner, drawn as the dart on the compass ring. While driving, the relative angle α (bearing minus own course, wrapped to ±180°) drives the blinker chevrons, the "at N o'clock" wording (each hour is 30 degrees), and the off-course warning. Course over ground comes from the GPS and is meaningful only at speed, which is why a stopped vehicle switches to compass wording: there is no nose to be relative to.

8.3 Dead reckoning between fixes

Radio updates arrive at most every ten seconds, but the screen renders twice per second, so between packets each vehicle's last fix is projected forward along its last known course and speed. The projection moves the latitude by the northward component of the traveled distance and the longitude by the eastward component, with the cosine of latitude correcting for the fact that lines of longitude converge toward the poles:

$$\varphi(t) = \varphi_0 + \frac{v\,\Delta t \cos\psi}{R}, \qquad \lambda(t) = \lambda_0 + \frac{v\,\Delta t \sin\psi}{R\cos\varphi}, \qquad \Delta t \le 120 \text{ s}$$

Projection is capped at 120 seconds, beyond which the estimate is no longer trustworthy. The age counter is deliberately not reset by projection: the motion on the screen stays smooth, while the age and the hollow dart tell the truth about how fresh the underlying measurement actually is.

8.4 Closing-speed trend

The trend label tells the driver whether the gap between the vehicles is shrinking or growing. It is driven by an exponential moving average of the rate of change of distance, sampled at every 500-millisecond render tick: each new sample contributes ten percent of the estimate while the history retains ninety percent, and a positive rate means the vehicles are closing:

$$r_k = 0.9\, r_{k-1} + 0.1 \cdot \frac{d_{k-1} - d_k}{\Delta t}, \qquad \Delta t = 0.5 \text{ s}; \quad r > 0: \text{ closing}, \quad r < 0: \text{ apart}$$

The 0.9 to 0.1 weighting settles in roughly five seconds, which provides enough smoothing that ordinary GPS jitter does not flap the label. The result is displayed as "closing N mph", "apart N mph", or "holding"

inside a 2 mph deadband, and it is suppressed entirely while the partner's data is stale, since a trend computed from a projection would be a guess about a guess.

9. Timing, liveness, and honesty semantics (Idea 4)

Nothing in a radio mesh can prove that a silent node is alive, so every liveness indicator in the system reduces to a comparison between the current time and a stored last-heard timestamp. The stock firmware's definition of an online node is exactly one such inequality:

$$\text{online}(n) \Leftrightarrow t_{now} - t_{last\ heard}(n) < 7200\ \text{s}$$

The 7200 s constant is stock firmware: `#define NUM_ONLINE_SECS (60 * 60 * 2)` in `src/mesh/NodeDB.cpp` of meshtastic/firmware [3], sized for an ecosystem where solar and battery nodes may legitimately stay silent for hours. The window must exceed the longest honest silence or healthy nodes flap offline; the cost is that a dead radio looks alive for up to two hours. Our traffic contract guarantees a packet at least every 300 s, so the display layers stricter thresholds:

Indicator	Threshold	Meaning
SF<->RG OK tag	heard ≤ 90 s	Link actively fresh
Hollow dart, inverted age	age > 30 s (partner moving) or > 90 s (partner parked)	Position projected by dead reckoning, not measured; threshold tracks the partner's broadcast cadence
DATA STALE banner	age > 120 s	Stop trusting the position
Radio OLED "online"	heard ≤ 7200 s	Stock ecosystem bound

Cadence design rule: every periodic packet names the failure it prevents. Position at 10 s / 20 m moving (tracking staleness), position at 60 s parked (presence), telemetry at 300 s (counter lies), NodeInfo on boot and periodically (names and keys).

10. Build, flash, and test procedures

The commands below build the firmware, flash it to a display, open a serial monitor, and run the navigation unit tests on the host computer. Two build environments exist: the real Bluetooth build for hardware, and a simulation build that feeds the interface fake data so the screens can be exercised on a desk without any radios. The notes that follow the commands record the environment-specific traps this project has already paid for.

```
# build + flash (ALWAYS pass --upload-port; check the SER= MAC first, Section 2)
pio run -e display-ble -t upload --upload-port COMx   # real BLE build
pio run -e display     -t upload --upload-port COMx   # SIM build: fake data, BOOT cycles
pio device monitor --port COMx --baud 115200
pio test -e native                                    # 12 nav unit tests on the host
```

- Board flags matter: pioarduino platform, `memory_type dio_opi`, flash mode `dio`. QIO silently boot-loops this panel. Do not "fix" it back.
- On the project workstation the native tests need MinGW on PATH: `PATH=/c/mingw64/bin:$PATH pio test -e native`.
- Every serial connection to a radio or display hard-resets it via DTR. Consequences: config writes can silently revert if the port closes mid-save (verify by read-back AND OLED), RAM-only node-table state is lost on each connect, and orphaned monitor processes hold COM ports (kill `device monitor` processes before flashing).
- Bring-up of a new display follows the checklist in `firmware-display/README.md`: SIM proof, BLE link, partner tracking, the position-request test (partner parked, age ≥ 30 s, press BOOT, expect the age to snap to seconds), and cased-versus-bare BLE RSSI within 2 dB.

11. Verification record

Claim	Evidence
Two-way link on Group9	Round-trip traceroutes across multiple sessions: 5.75 to 7.25 dB SNR per direction
Position agreement at rest	Two independent fixes 13 to 45 ft apart on one table (consumer GNSS scatter)
Live tracking	Display tracked a real drive on first boot; the 10 s moving cadence was proven in the Week 4 drive test
On-demand position request	CLI and display-button requests both returned the partner's live fix at full precision
Messaging	Canned texts delivered both directions; popup confirmed receiver-side only on hardware
Airtime claim	CLIENT_MUTE halved measured idle utilization from 9-13 to 5-6 percent
Unit tests	12/12 navigation-math tests pass on the native host build
Two-vehicle drive test (2026-10-04)	Full convoy session on video: live tracking at 17 mph with blinkers and the off-course banner, separation to 0.5 mi and reunion ("closing 13 mph"), messaging both ways, one continuous session with zero reconnects
Environment limits observed	DATA STALE onset near 0.5 mi in a built-up area with in-cabin antennas: an environment number, not a range number; consistent with the 20 dB reserve being consumed by car body and buildings
Open-road range test (2026-10-05)	Full two-way tracking held to 3.26 mi (5.25 km): LINK CONNECTED with both coordinates on screen and the displayed distance matching the coordinate delta exactly. Stale edge by 3.75 mi (6.03 km), flagged honestly on both devices. Design range of 3 to 5 km met at the top and exceeded by about a kilometer; full data in `docs/RANGE_TEST_RESULTS.md`
Guided arrival (filmed)	On the return leg the display read roughly 700 ft at 10 o'clock, the driver turned to follow, the partner's vehicle entered the windshield view, and the session ended at 25 ft, stopped: the clock-direction guidance verified end to end against ground truth

The two drive tests closed the loop twice over. The 10-04 night drive produced the course-up compass and the cadence-aware staleness cue of Section 7.4, shipped the next day; the 10-05 range test then validated

both on the road (dart at 12 o'clock at 37 mph; a parked partner at age 1m displayed calm while a true 2-minute outage displayed inverted and bannered) and converted the Section 4 range prediction into a measurement. In-town versus open-road stale onset, 0.5 mi against 3.75 mi on identical hardware, is the 20 dB environmental reserve made visible. The remaining optional refinement is a repeat with a roof-mounted antenna to quantify the car-body penalty directly.

12. Failure modes and recovery (the scar catalog)

Every entry below happened on this hardware and was reproduced or root-caused before being written down.

#	Symptom	Root cause	Recovery
1	Total mutual silence; configs read back identical; 0 percent channel utilization	Frequency slot mismatch: a channel write silently reverted, leaving the radios on different slots with the same key (Ideas 1 and 2)	Check both OLED LoRa screens: Group9 must read 912.875. Re-apply the channel URL, verify, power-cycle, verify again
2	Battery ran flat; the radio later boots looking healthy but config is factory or channels are lost	Battery-death crash reverts NVS configuration (observed: one radio lost channels only, one lost everything including keys and name)	Full provisioning recipe. Pre-flight rule: never run radios to empty; check 912.875 before every session
3	Screen lit, GPS fix shown, but the peer hears nothing for hours and USB serial refuses to connect	Half-alive wedge after a flat battery: the UI runs while LoRa TX and serial are frozen	Full power-cycle. Judge a radio by the OTHER radio's last-heard, never by its own screen
4	Broadcasts flow but direct messages and position requests time out in one direction, with no NAK	Stale public key: the peer's key regenerated in a factory event; the requester encrypts to the old key; the receiver drops silently	`--reset-nodedb` on the requester, wait one minute for NodeInfo relearn, retry
5	"Online" counters lie in either direction; entries stamped months old; distance shows kilometers between adjacent radios	Stale stored timestamps and positions: fixless radios re-broadcasting old fixes, flash-reloaded node records, clock-less boots (Idea 4)	Purge stored positions (`--remove-position`), sync clocks, reset node DBs. The 300 s heartbeat prevents recurrence
6	A config write appears to succeed; later reads show the old value	The serial port-close DTR reset rebooted the chip mid-save	Always verify by read-back AND the OLED; for stubborn values use the Python API holding the port open ~20 s
7	Node table shows nameless ghost entries; freshly learned entries vanish after a USB session	Crash-corrupted records; RAM state wiped by the connect reset before the scheduled flash save	`--reset-nodedb`, then leave the radio off USB; entries repopulate from live traffic
8	A Meshtastic radio erased by a firmware upload	An auto-picked COM port hit the radio (shared VID:PID)	esptool erase, factory `firmware-heltec-v4` binary at 0x0, then full provisioning; the node number survives (MAC-derived)

Appendix A: acronyms and abbreviations

Term	Expansion and meaning in this system
AES-256	Advanced Encryption Standard, 256-bit key; encrypts all channel traffic
BLE	Bluetooth Low Energy; the in-cab display-to-radio link at 2.4 GHz
BW	Bandwidth; RF channel width, 250 kHz here
CAD	Channel activity detection; the SX1262's listen-before-talk preamble sniff (Section 4.3)
CSMA	Carrier-sense multiple access; the listen-then-randomly-back-off channel discipline (Section 4.3)
CFR	Code of Federal Regulations; 47 CFR 15.247 governs this band
ChUtil	Channel utilization; percent of the last minute the channel carried RF energy
CR	Coding rate; forward-error-correction ratio, 4:5 here
CRC	Cyclic redundancy check; per-packet integrity checksum
dB / dBm / dBi	Decibel ratio / power referenced to 1 mW / antenna gain referenced to an isotropic radiator
DM	Direct message; addressed to one node, per-node encrypted
DTR	Data Terminal Ready; serial control line whose toggle hard-resets the ESP32
EMA	Exponential moving average; smooths the closing-speed trend
FCC	Federal Communications Commission; US radio regulator
FSPL	Free-space path loss; the distance term of the link budget
GATT	Generic Attribute Profile; BLE's service and characteristic structure
GNSS / GPS	Global navigation satellite system / the US constellation specifically
ISM	Industrial, Scientific, Medical; the license-free 902 to 928 MHz band
LoRa	Long Range; chirp spread spectrum modulation
MCU	Microcontroller unit; the ESP32-S3 in every device here
NAK / ACK	Negative acknowledgment / acknowledgment; delivery failed / delivered
NVS	Non-volatile storage; the ESP32 flash key-value store holding configuration
OLED	Organic LED; the radio's small status screen
PHY	Physical layer; the radio chip and modulation level of the stack
PKC	Public-key cryptography; Curve25519 pairs used for direct messages
PSK	Pre-shared key; the channel secret both radios hold
RLCD	Reflective liquid crystal display; sunlight-readable, backlight-free panel
RSSI	Received signal strength indicator; raw received power
SF	Spreading factor (context: modulation); also the callsign of radio 1
SNR	Signal-to-noise ratio; signal height above the noise floor, in dB

SMA	SubMiniature version A; the screw-on antenna connector
ToA	Time on air; the channel time one packet occupies
UUID	Universally unique identifier; names GATT services and characteristics
UTC	Coordinated Universal Time; the GPS-supplied wall clock

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